Ming Li

PhD Candidate, Computer Science, University of Maryland Expected 2027

mingliiii.github.io | Google Scholar | GitHub | LinkedIn | minglii@umd.edu

2026 Apple Scholar in AI/ML | **2× ACL Oral (2025, 2026)** | **ACL'26 SAC Highlight**

Best Research, e-Assessment Awards 2026 | **Best Paper Honorable Mention, NeurIPS'26 AI-Native Academia Workshop**

I study how to train, understand, and rigorously evaluate LLMs and agentic systems, with a focus on post-training, reasoning behavior, and reliable AI evaluation.

SELECTED RESEARCH

Data-Centric LLM Post-Training

Data selection and curation for efficient LLM post-training, with gradient-based analysis of data quality.

- **IFD / Cherry LLM** — NAACL'24. Introduced instruction-following difficulty for model-specific, self-guided data selection; outperformed full-data instruction-tuning baselines using only 5–10% of the training data. **500+ citations**.
- **Data Quality through Layer-wise Gradients** — ACL'26. Unified disparate data-quality metrics through layer-wise gradient spectra, providing a common view of how instruction and reasoning data shape post-training. **AC-recommended Oral · Top-15% meta-review**.

Reasoning & Model Behavior

LLM interpretability and model-behavior analysis through gradients, token dynamics, and reasoning traces.

- **Schoenfeld's Anatomy / ThinkARM** — **ACL'26 Oral & SAC Highlight**. Developed a framework grounded in cognitive science and Schoenfeld's Episode Theory to map reasoning traces into functional episodes and analyze reasoning structure beyond token-level statistics.
- **Fast vs. Slow Thinking** — **ACL'25 Oral**. Studied how reasoning traces shape layer-wise gradients during post-training, showing that training on detailed reasoning produces smaller, more stable gradients than direct-answer training.

AI Evaluation & Reliability

Stress-testing models and agents to uncover capability limits, behavioral failures, and unreliable judgments.

- **Moltbook / AI Agent Society** — ACM CAIS'26. Diagnosed multi-agent interaction dynamics, finding persistent individual inertia without stable collective consensus; **Science News interview**.
- **RobustReview / SciCore** — NeurIPS'26 AI-Native Academia Workshop; **Best Paper HM & Oral**. Formulated rhetorical robustness as a missing dimension of trustworthy AI review: stable judgments under content-preserving rewrites without losing discrimination across papers; introduced SciCore to improve this balance.

INDUSTRY & VISITING RESEARCH EXPERIENCE

MBZUAI — Visiting Researcher Feb–May 2026

Rhetoric Reward-Hacking (2026 preprint): Built and ran a **large-scale LLM rewriting-and-evaluation pipeline over 4,200 full manuscripts and 42K+ AI reviews**, orchestrating multiple LLM APIs for controlled, recursive, and reviewer-guided interventions while enforcing content-preservation constraints; exposed LLM-as-a-judge sensitivity to evidence framing and novelty claims.

Amazon — Applied Scientist Intern May–Aug 2025

RLAAR (ACL'26): Developed **curriculum reinforcement learning with verifiable rewards (RLVR)** for multi-turn reliability by **extending verl to support fully dynamic multi-turn on-policy rollouts under GRPO** and implementing mixed verifiable accuracy/abstention rewards; improved Qwen3-8B Lost-in-Conversation retention from 62.6% to 75.1%.

Microsoft Research — Research Intern Feb–May 2025

FoReaL-Decoding (NeurIPS'25 Efficient Reasoning Workshop): Token-level analysis revealed reasoning/non-reasoning LLM divergence at sentence-opening "thinking cues." Implemented a **custom collaborative decoder over Hugging Face Transformers** for efficient inference, reducing theoretical FLOPs by 30–50% and chain-of-thought (CoT) length by up to 40%.

Adobe Research — Research Scientist/Engineer Intern May–Nov 2024

TRIG (ICLR'25 Reasoning and Planning for LLMs Workshop): Introduced the first benchmark and instruction-tuning dataset for text-rich document QA grounding. **Built a synthetic data pipeline from raw documents to ∼96K training examples via multimodal alignment, iterative refinement, and filtering**; implemented distributed SFT of LLaVA-13B and MLLM evaluation.

Ming Li

SELECTED PUBLICATIONS

Selected publications and manuscripts · 20+ first/co-first-author papers overall

* Equal contribution

Full list: Homepage | Google Scholar

Data-Centric LLM Post-Training

1K+ GitHub stars across these post-training codebases

From Quantity to Quality: Boosting LLM Performance with Self-Guided Data Selection for Instruction Tuning.
M. Li, Y. Zhang, Z. Li, et al. — NAACL 2024

Superfiltering: Weak-to-Strong Data Filtering for Fast Instruction-Tuning.
M. Li, Y. Zhang, S. He, et al. — ACL 2024

How Instruction and Reasoning Data Shape Post-Training: Data Quality through the Lens of Layer-wise Gradients.
M. Li, Y. Li, Z. Li, T. Zhou — ACL 2026; **AC-recommended Oral · Top-15% meta-review**

Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.
M. Li, P. Chen, Z. Zhang, et al. — ACL 2026

ATLaS: Agent Tuning via Learning Critical Steps.
Z. Chen*, **M. Li***, Y. Huang, et al. — Findings of ACL 2025

Reasoning & Model Behavior

Schoenfeld's Anatomy of Mathematical Reasoning by Language Models.
M. Li*, C. Fan*, Y. Cheng*, S. Feizi, T. Zhou — **ACL 2026 Oral & SAC Highlight**

What Happened in LLMs Layers when Trained for Fast vs. Slow Thinking: A Gradient Perspective.
M. Li, Y. Li, T. Zhou — **ACL 2025 Oral**

Missing Premise Exacerbates Overthinking: Are Reasoning Models Losing Critical Thinking Skill?
C. Fan*, **M. Li***, L. Sun, T. Zhou — COLM 2025

What Does RL Improve for Visual Reasoning? A Frankenstein-Style Analysis.
X. Li*, **M. Li***, T. Zhou — COLM 2026

AI Evaluation & Reliability

Does Socialization Emerge in AI Agent Society? A Case Study of Moltbook.
M. Li*, X. Li*, T. Zhou — ACM CAIS 2026

ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness.
Y. Liang*, **M. Li***, C. Fan, et al. — NeurIPS 2025 Datasets & Benchmarks

BenTo: Benchmark Reduction with In-Context Transferability.
H. Zhao, **M. Li**, L. Sun, T. Zhou — ICLR 2025

Rethinking LLM-as-a-Judge: Representation-as-a-Judge with Small Language Models via Semantic Capacity Asymmetry.
Z. Li, Y. Zhang, **M. Li**, et al. — ICLR 2026

EDUCATION

University of Maryland — PhD Candidate, Computer Science — 2023–Present
Advisor: Tianyi Zhou | Earliest graduation: Spring 2027 · Graduation timing flexible

Texas A&M University — MS, Computer Science — 2021–2023
Advisor: Ruihong Huang

Xi'an Jiaotong University — BS, Computer Science — 2016–2020

RESEARCH LEADERSHIP, HONORS & GRANTS

Additional Honors & Grants. Tencent Project Up Student Travel Grant, COLM 2026 (up to RMB 40,000 / US$6,000) · UMD Graduate School Dean's Fellowship (2023) · UMD Jacob K. Goldhaber Travel Grant (2024).

Community & Outreach. ACL ARR Area Chair · AI/ML technical content creator, **2.8K followers** on RedNote (Xiaohongshu) · CIKM'25 tutorial contributor.

Mentorship. Mentored junior researchers across multiple first-author projects, from problem formulation through publication and independent ownership.

Technical Skills. PyTorch · Hugging Face Transformers · verl · vLLM · distributed training · supervised fine-tuning (SFT) · reinforcement learning (RLHF/RLVR) · PPO · DPO · GRPO · on-policy distillation (OPD) · reward design · synthetic data · multimodal learning · LLM/VLM evaluation.